

Research paper

Enhanced perturb and observe control for addressing power loss under rapid load changes using a buck–boost converter

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ABSTRACT

Many researchers have focused on improving the efficiency of photovoltaic (PV) systems by enhancing their control mechanisms to more consistently operate in the PV panels' maximum power points. The predominant method used in these studies is the Perturb and Observe (P&O) method. However, this method faces challenges of drift when encountering rapid changes in resistive load, which completely inhibits them from addressing the problem of power loss. Additionally, the traditional P&O method suffers from oscillation issues. Here, an innovative control method is presented to improve the duty cycle and enable the handling of the challenges associated with sudden and rapid changes in resistive loads. A buck–boost power converter is implemented as an impedance adaptor to maintain operation at the maximum power point. The efficiency of the proposed modified P&O algorithm was evaluated using MATLAB software and compared to the traditional P&O algorithm across various operating scenarios. The simulation results show that the proposed method is significantly more efficient and effective than the traditional P&O algorithm, indicating its potential to enhance the efficiency of PV systems and increase their ability to adapt to changes in resistive load.

1. Introduction

Humans are currently facing an environmental crisis caused by air pollution that is associated with toxic gases that are emitted as a result of combusting petroleum and natural gas. This pollution represents one of the most prominent challenges affecting the environment, human health, animals, and plants (Benbouhenni et al., 2022a). Also, the high demand for electrical energy (EE) has led to high consumption of traditional sources causing toxic gases, which is undesirable. Therefore, the high demand for EE contributes significantly to the phenomenon of global warming, which requires thinking about effective and appropriate solutions to overcome this problem, which has become a major threat to the environment and human life. Recently, unknown diseases have appeared for which there is no cure due to the presence of toxic gases. It has become necessary to think seriously about reducing these toxic gases, and this is by creating effective solutions for generating EE.

Scientific studies have shown that toxic gases emitted from the burning of fossil fuels cause negative changes in air composition and have harmful effects on public health (Kadi et al., 2022). For example, carbon dioxide (CO₂) is considered one of the most prevalent toxic gases in the air that is generated due to the combustion of fossil fuels. This gas significantly contributes to the phenomenon of global warming, as it enhances the trapping of heat in the Earth's atmosphere, leading to an increase in global temperature (Benbouhenni et al., 2022b). The negative effects of global temperature increases on the environment and public health are extensive, leading to the melting of glaciers, rising sea levels, and threats to coastal areas and communities. Additionally, air pollution with toxic gases also contributes to the formation of acid rain, which reacts with water to form toxic gases in the atmosphere, causing chemical changes to the natural environment, such as increased acidity of surface water and soil, which affects marine and plant life (Chojaa et al., 2023). Considering the detrimental environmental impacts and health challenges arising from air pollution, it is imperative to explore

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Nomenclature	
PV	Photovoltaic system
WE	Wind energy
RESs	Renewable energy sources
MPP	Maximum power point
AI	Artificial intelligence
VSSs	Variable step sizes
TOSMC	Third-order sliding mode control
FSCC	Fractional short circuit current
PSO	Particle swarm optimization
GWO	Grey wolf optimization
RE	Renewable energy
EE	Electrical energy
MPPT	Maximum power point tracking
P&O	Perturb and observe
SE	Solar energy
BBFB	Buck–boost full-bridge
PSC	Partial shading conditions
IPSO	Improved particle swarm optimization
GMPP	Global maximum power point
ANFIS	Adaptive neuro-fuzzy inference system

effective solutions to improve air quality and safeguard the health of both the environment and its inhabitants. Renewable energy sources (RESs), such as solar energy (SE) and wind energy (WE), are increasingly gaining popularity as a clean and sustainable alternative to traditional energy derived from fossil fuels (Gasmi et al., 2023a). These sources offer opportunities for energy generation at a lower cost and with reduced environmental impact, making them an ideal solution for alleviating environmental pressure and reducing air pollution. Consequently, the transition to utilizing RESs is deemed an urgent necessity for preserving human health and environmental sustainability in the future. Research in environmental science consistently demonstrates the adverse effects of air pollution on both ecosystems and human health (Gasmi et al., 2023b). Air pollutants, such as particulate matter, nitrogen oxides, sulfur dioxide, and volatile organic compounds, can lead to respiratory diseases, cardiovascular problems, and even premature death. Additionally, air pollution contributes to environmental degradation, including acid rain, smog formation, and ecosystem disruption. Moreover, public health studies emphasize the importance of mitigating air pollution to reduce the burden of disease and improve overall well-being. Transitioning to RESs is recognized as a crucial strategy for addressing this issue. By reducing reliance on fossil fuels and promoting cleaner alternatives such as SE and WE, societies can significantly mitigate air pollution and its associated health risks (Gasmi et al., 2022). The use of RESs contributes significantly to overcoming the increasing demand for EE, which is a positive matter, as the use of these energy sources allows the individual to obtain EE, which reduces the pressure on the electrical network. Furthermore, advancements in renewable energy (RE) technologies have made them increasingly accessible and cost-effective. Innovations in solar panel efficiency, wind turbine design, and energy storage solutions have reduced the costs of RE production, making it a viable option for widespread adoption. The transition to RESs is essential for addressing the environmental and health challenges associated with air pollution and for ensuring the long-term sustainability of our planet (Kadi et al., 2023). Comparing WE and SE reveals that relying on SE is a cost-effective option (Yesssef et al., 2024). In the work (Angadi et al., 2021), the author used three different systems to generate EE and to overcome the increasing EE demand. These energy systems are the SE system, WE system, and thermal power systems, where EE is generated from several different sources, which makes this system effective. The use of integration between three different systems

makes controlling the system complex and expensive, which is undesirable. Also, the use of three different sources makes it difficult to implement this system on the ground. This hybrid energy system was verified using simulation, and the results showed that the proposed system is effective and reliable in generating EE in various difficult conditions, which is a positive thing. Relying on RESs alone to generate EE causes several problems, especially in difficult weather conditions, which is undesirable and leads to interruptions in EE. Therefore, the author of the work (Merabet et al., 2017) proposed an energy management and control system for microgrids based on hybrid energy resources such as SE, WE, and battery. The battery was used as a suitable solution for this hybrid system to store energy and use it when needed or in the case of difficult weather conditions. This proposed hybrid energy system relies on the use of a control algorithm to operate its various sections well while avoiding any possible malfunction. The advantage of this proposed energy system is that it generates and stores energy, making it one of the most prominent systems that can be relied upon in the future. The experimental results obtained using dSPACE 1104 highlighted the effectiveness and performance of this proposed system in difficult conditions, which makes this system of great importance at the public and private levels. This system has disadvantages such as complexity of control, high costs, difficulty of control, and a battery protection system that requires the use of an algorithm characterized by high performance because the battery is the weakest link in this studied energy system. Leveraging SE includes individuals and governments installing photovoltaic (PV) panels and connecting them to provide the necessary energy (Debdouche et al., 2023a). This technology is widely available in most regions, and solar panels can be easily installed on the surface roofs of homes and institutions. SE relies on converting sunlight into EE through the use of solar cells. This process is sustainable and does not produce greenhouse gas emissions, contributing to environmental preservation (Debdouche et al., 2023a). Additionally, the cost of installing and maintaining SE systems significantly decreases over time. Solar panels are among the most prominent technologies used to generate clean materials and REs. Given their significance, researchers and scientists are diligently working on further developing this technology and improving its performance (Debdouche et al., 2023b). The focus in this context revolves around several scientific aspects to enhance the efficiency of solar panels. A key challenge facing solar panels is the development of advanced control systems, which contribute to improving their efficiency by capturing the maximum power point tracking (MPPT) and reducing fluctuations around it (Xu et al., 2021). This improvement increases the efficiency of converting SE into EE effectively and sustainably. In the work (Walid et al., 2023), the author used a nonlinear MPPT strategy based on the use of third-order sliding mode control (TOSMC) to control the PV system. This proposed strategy is characterized by high robustness and its ability to significantly improve the system characteristics compared to the traditional MPPT strategy. The MPPT-TOSMC strategy is a strategy different from the traditional MPPT strategy, as it was proposed to use the TOSMC strategy because of its many advantages, including high performance, durability, ease of implementation, a small number of gains, and its lack of need for a mathematical model of the system, which makes it the most suitable, especially in the event of a malfunction in the system. The MATLAB environment was used to implement the MPPT-TOSMC strategy and compare it with the traditional strategy, and several tests were used for this purpose. Simulation results show the high performance of the MPPT-TOSMC strategy in improving the characteristics of the PV system compared to the traditional strategy. In the work (Salman et al., 2018), the author used a perturb and observe (P&O) algorithm-based MPPT technique to control a 200 W PV system, where a battery was used to store energy. Using the P&O strategy allows for a significant increase in the characteristics of the MPPT strategy, which leads to increasing the performance and durability of the PV system. The proposed MPPT-P&O strategy is characterized by simplicity, ease of implementation, high durability, and great efficiency, which makes it

the most appropriate and one of the best solutions that can be relied upon. This proposed strategy was implemented experimentally and its behavior and performance were compared with the traditional strategy, where a microcontroller was used for this purpose. The experimental results show the high performance of the proposed strategy in improving the characteristics of the studied energy system compared to the traditional strategy. Grey wolf optimization (GWO) technique is the strategy that was proposed in work (Mohanty et al., 2016) to overcome the drawbacks of MPPT of PV systems under partial shading conditions (PSCs). The GWO is a new optimization method that overcomes the limitations such as lower tracking efficiency, steady-state oscillations, and transients. The proposed strategy is characterized by high performance and great durability, as the GWO strategy was used to determine the values of gains, which allows for obtaining excellent results compared to both P&O and improved particle swarm optimization (IPSO) techniques. The MPPT-GWO strategy was implemented in the MATLAB environment, comparing the behavior with traditional strategies, and several different tests were used for this purpose. Also, the performance of the proposed strategy was verified using experimental work, where the experimental results showed that the proposed strategy has better performance and durability than P&O and IPSO-MPPTs. In the work (Sher et al., 2015) the author presented a new method for MPPT technique for PV systems, where the proposed method improves the operation of the P&O method in variable environmental conditions using the fractional short circuit current (FSCC) method. The positivity of the designed strategy lies in rapid tracking under changing environmental conditions. In addition, this scheme offers low-power oscillations around maximum power point (MPP) and, therefore, more power harvesting compared with the common P&O method. The performance of this proposed method was verified experimentally, where the dSPACE DS1104 was used to study the behavior and robustness of the proposed technique compared to the conventional technique. This designed technique provided satisfactory results in all completed tests, which were positive. The use of artificial intelligence (AI) techniques to overcome the defects and problems of the MPPT strategy in the SE system is considered one of the most reliable solutions (Yap et al., 2020), as the use of traditional MPPT techniques is unable to track the global MPP (GMPP) under partial shading condition. Integrating AI into the MPPT technique is critical to ensuring GMPP traceability while increasing the efficiency and overall performance of the MPPT technique. These proposed strategies have been implemented in a MATLAB environment, where all AI-based MPPT techniques show fast convergence speed, lower steady-state oscillation, and high efficiency, compared with conventional MPPT techniques. Using AI techniques is complex because each technology has its advantages and disadvantages, and these strategies rely heavily on experience. AI-based MPPT techniques are computationally intensive and expensive to achieve, which is a downside. So, a hybrid MPPT strategy is suitable in terms of balance between performance and complexity and combines the advantages of traditional and AI-based MPPT techniques. To enhance the conversion of PV power generation under fluctuating sun irradiance, the author in the work (Priyadarshi et al., 2020) proposed a new MPPT strategy based on adaptive neuro-fuzzy inference system-particle swarm optimization (ANFIS-PSO) to obtain fast and maximum PV power with zero oscillation tracking. The MPPT-ANFIS-PSO method has no extra sensor requirement for the measurement of irradiance and temperature variables. The proposed MPPT-ANFIS-PSO strategy was implemented using simulation and experimental work, where the dSPACE interface was used to implement it experimentally, comparing the performance with the strategies P&O technique, PSO strategy, ant colony optimization, and artificial bee colony MPPT methods for the PV system. The obtained experimental results demonstrate the simulated results and the extent of the high performance of the MPPT-ANFIS-PSO strategy in improving the dynamic response of the studied energy system compared to other strategies. The negative of this proposed strategy lies in the use of the neuro-fuzzy algorithm, as there is no mathematical rule that facilitates

its application in order to obtain excellent results, as it relies heavily on experience. In (Swaminathan et al., 2022), the author used the fixed zone P&O-MPPT technique as a suitable solution to control the PV system. This proposed strategy is different from the traditional MPPT strategy in terms of principle, ease of implementation, durability, performance, and efficiency. This proposed strategy was used to overcome the disadvantages and problems of the PV system, such as low current and power quality. The performance and robustness of the proposed strategy are validated through a comprehensive set of hardware experiments on a buck-boost full-bridge (BBFB) converter under different radiation conditions according to the EN50530 standard, where the performance was compared with both conventional and variable step sizes (VSS) techniques. To obtain a gradual change in irradiance, it is experimentally verified that the proposed strategy is 42 % and 20 % more efficient than conventional and VSS techniques, respectively. During variable radiation conditions using the proposed strategy, the maximum power loss is one-sixth compared to conventional and VSS techniques. Type 2 fuzzy logic control (T2FLC) is the proposed strategy to overcome defects and problems of the MPPT strategy in (Maroua et al., 2024), where this proposed technique is used to control the microgrid-connected PV system with battery energy storage through multi-functional voltage source inverter using direct power control. The strategy used depends largely on experience, as its use does not require precise knowledge of the mathematical model of the system, which makes it give excellent results in the event of a malfunction in the system. This strategy has also been used to control DC link voltage. So the MPPT strategy based on the T2FLC controller is a strategy different from the conventional technique, where robustness and high performance are among the most prominent features, and this is shown by the results of the simulation carried out using the MATLAB environment. The negative of this proposed strategy lies in the number of fuzzy logic rules that must be adopted to obtain good results, as the experimentation method is used to determine the best number of rules that must be relied upon. In the work (Boutaghane et al., 2024), the author relied on fractional-order integral sliding mode controls (SMCs) as a suitable solution to control the three-phase grid-connected PV inverter system. This proposed strategy relies on the combination of fractional calculus and SMC, where high performance, efficiency, and durability are among the most prominent features of this proposed MPPT strategy. The latter was used to increase the characteristics of the MPPT strategy of the PV system, as the simulation results in all tests showed that the use of this strategy led to an increase in the characteristics of the MPPT strategy, and this appears through improving the quality of power and current compared to the traditional MPPT strategy. The negative of this proposed MPPT strategy lies in its use of the mathematical model of the system, which is a negative that contributes to raising ripples if the system parameters change. Also, using the SMC strategy leads to the problem of chattering, which is undesirable. In the work (Naamane et al., 2024), the author used the integral robust nonlinear backstepping command (IRNBC) to improve the characteristics of the PV system, and this strategy was used to increase the performance and robustness of the MPPT strategy. The use of this nonlinear strategy allows for a significant improvement in the quality of power generated by the PV system compared to the use of the traditional MPPT strategy, and this is shown by the simulated results in all tests performed, where a significant improvement is observed in the values of both ripples, harmonic distortion of current, and overshoot of power. Complexity and difficulty of implementation are among the most prominent drawbacks of this proposed MPPT-IRNBC strategy compared to the traditional MPPT strategy. Also, the use of the backstepping control strategy makes the proposed MPPT-IRNBC control linked to the mathematical model of the system under study, which is undesirable and makes the power quality low in the event of a malfunction. In the work (Rezkaallah et al., 2018), the author improved the P&O algorithm using SMC to control a hybrid system consisting of a PV system and a wind system. The proposed MPPT-SMC-P&O strategy was used to increase the robustness and performance of the MPPT strategy of the PV system,

where the behavior and performance were verified using simulation and DSP-dSPACE. Experimental and simulation results show the superiority of the proposed MPPT-SMC-P&O strategy over the traditional MPPT strategy in terms of improving the characteristics of the studied hybrid system. The proposed MPPT-SMC-P&O strategy has drawbacks, including complexity, difficulty of implementation, and reliance on the mathematical model of the studied system, which is undesirable and contributes to the deterioration of power quality in the event of a malfunction in the system. Additionally, researchers are focusing on improving the response time of solar panels to ensure their effective response to rapid fluctuations in solar irradiation and temperature. They are working on developing techniques that enable solar panels to adapt quickly to changes in the external environment, thereby improving their performance and increasing their ability to generate energy with high efficiency. Furthermore, it is essential to address the drift in searching for the MPPT technique of solar panels, especially under rapid and unexpected load changes (Lu et al., 2018). Researchers are also working on developing artificial intelligence systems and machine learning techniques that enable solar systems to adapt quickly to sudden load changes, thus maintaining their efficiency and sustainability in continuously generating SE effectively. In summary, efforts are ongoing to improve the performance of solar panels by developing controls and fast response techniques to enhance their efficiency and sustainability as reliable sources of clean materials and REs in the future. Here, we introduce substantial improvements to the PV-SE system through the development of a new P&O control model. We transformed the P&O-MPPT technique from a weak classical control system to a new P&O-MPPT model with advanced technological capabilities, enabling it to effectively address multiple technical aspects, including:

1. Eliminating the drift problem: By refining the algorithm of the P&O-MPPT method, we successfully eliminated the drift issue that previously hindered the accurate tracking of the MPP of solar panels.
2. Addressing oscillations: Thanks to the new P&O-MPPT technique, we managed to mitigate oscillations resulting from fluctuations in solar irradiance conditions, ensuring greater stability in the performance of the P&O-MPPT system.
3. Maximizing MPPT technique: Through the application of novel techniques in the algorithm, we enhanced the system's ability to achieve the maximum available power from solar panels at all times.

These enhancements offer a financial advantage by reducing wasted energy costs and improving system efficiency. Additionally, they contribute to enhancing the economic performance of solar projects and increasing return on investment.

The remainder of this manuscript is organized as follows. The components of the PV system are discussed in the second section, where the PV panel, buck-boost converter, P&O algorithm, and drift and power loss problems are discussed. In the third section, the proposed solution is discussed to address the oscillation problem. The fourth section proposes a solution to address the drift problem. Simulation results generated in MATLAB software are listed in Section V. Conclusions of the work are provided in Section VI.

2. The PV system components

The PV system comprises PV panels, a DC-DC converter, an ESS, and a load. The PV panels are considered the main component of the system, converting solar energy into EE, while the converter transforms EE. Additionally, the ESS stores excess energy generated during times of surplus production, and the load system includes devices that efficiently consume the generated EE. Understanding these components and their interactions is crucial for significantly improving the efficiency and performance of PV systems, thereby enhancing the sustainability and effectiveness of RE utilization.

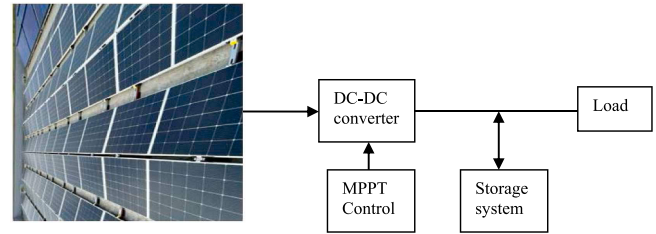


Fig. 1. PV system components.

2.1. PV panel

A solar panel is a device used to convert SE received from the sun into EE. It comprises solar cells made of semiconductor materials, and these cells are also known as PV cells (Fathabadi, 2019). The function of these cells is to absorb sunlight and convert SE into an electric current that can be used in EE devices.

The electrical model of a solar cell is represented in Fig. 2. This model consists of several electrical components that simulate the behavior of a solar cell under varying conditions of solar irradiance and temperature. The model typically includes components such as a current source representing the generated photocurrent, a diode representing the solar cell's behavior as a P–N junction, a series resistor representing the cell's internal resistance, and a parallel resistor representing the shunt resistance (Mahmud et al., 2017). These components are interconnected according to the physical principles governing the operation of the solar cell, and the model is used to analyze and predict the performance of the solar cell under different operating conditions.

A PV current source refers to a component or device within a PV system that generates a direct electric current (DC) when exposed to sunlight. This current is produced through the PV effect, where semiconductor materials within the device absorb photons from sunlight, creating electron-hole pairs and generating an electrical current (Gulzar et al., 2023). The PV current source is a fundamental element in SE systems, as it converts SE into usable EE.

I_{PV} is formulated as:

$$I_{PV} = I_{ph} - I_d - I_{sh} \quad (1)$$

$$I_{PV} = I_{ph} - I_0 \left[\exp \left(\frac{qV_{PV} + qR_s I_{PV}}{N_s K T n} \right) - 1 \right] - \frac{V_{PV} + R_s I_{PV}}{R_{sh}} \quad (2)$$

where, V_{PV} is the PV voltage, I_{PV} is the output current, I_{ph} is the photocurrent generated by the cell, I_0 is the reverse saturation current of the cell, V_{PV} is the output voltage, R_s is the series resistance, q is the elementary charge, n is the ideality factor, k is the Boltzmann constant, and T is the absolute temperature in Kelvin.

I_{ph} is formulated as:

$$I_{ph} = [I_{SC} + K_{SC}(T - 298)] \frac{G}{G_{SC}} \quad (3)$$

where, I_{SC} is the short-circuit current under standard test conditions, T is the cell temperature at 25 °C, $G(W/m^2)$ is the solar irradiation on the cell surface, and K_{SC} is the short-circuit current coefficient.

I_0 is influenced by temperature according to the following equation:

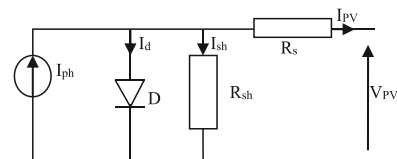


Fig. 2. Electrical model of a solar cell.

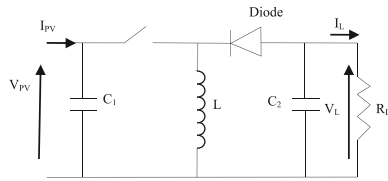


Fig. 3. Buck-boost converter.

$$I_o = \frac{I_{SC} + K_{SC}(T - 298)}{\exp[V_{OC} + K_V(T - 298)/a \times V_{th}] - 1} \quad (4)$$

V_{th} is the thermal voltage of the cell, which is expressed as:

$$V_{th} = (K \times T)/q \quad (5)$$

2.2. Buck-boost converter

A buck-boost converter is a type of DC-DC converter that can both increase (increase) and decrease (decrease) the input voltage to provide a regulated output voltage (Andrade et al., 2022; Shi et al., 2011; Ado et al., 2020). This converter topology combines the functionalities of both the boost converter and the buck converter into a single circuit (Sangwongwanich et al., 2016). It is capable of efficiently converting a wide range of input voltages to a desired output voltage level, making it useful in various applications where voltage regulation is needed.

The input and output voltages are provided as follows:

$$\frac{V_L}{V_{PV}} = \frac{D}{1 - D} \quad (6)$$

The relationship between the input and output powers in an ideal (lossless) converter can be characterized by the following equations:

Let us denote the input power as P_{in} and the output power as P_{out} . In an ideal converter, where losses are negligible, the input power is equal to the output power. This can be mathematically expressed as:

$$P_{in} = P_{out} \quad (7)$$

$$V_{PV} \times I_{PV} = V_L \times I_L \quad (8)$$

$$\frac{V_L}{V_{PV}} = \frac{I_{PV}}{I_L} \quad (9)$$

$$\frac{I_{PV}}{I_L} = \frac{D}{1 - D} \quad (10)$$

In the context of a PV system, the input impedance R_{in} is a crucial parameter that affects the overall performance and efficiency of the system. It represents the impedance that the PV panel perceives from the external circuit connected to it. This impedance includes the internal resistance of the PV panel itself as well as the impedance of any external components or loads connected to it:

$$R_{in} = \frac{V_{PV}}{I_{PV}} \quad (11)$$

Substituting Eqs. (6) and (10) into Eq. (11) yields the formulation of the input impedance of the DC-DC converter (Mokaribolhassan et al., 2023):

$$R_{in} = \frac{(1 - D)^2}{D^2} \times \frac{V_L}{I_L} = \frac{(1 - D)^2}{D^2} \times R_L \quad (12)$$

Eq. (12) demonstrates that by adjusting the duty cycle (D), the input impedance (R_{in}) can be adjusted to match the optimal input impedance (R_{opt}) of the PV panel at the MPPT for R_L , where:

$$R_{opt} = \frac{V_{MPP}}{I_{MPP}} \quad (13)$$

2.3. P&O algorithm

The P&O algorithm is a method of controlling MPPT in SE systems. This approach utilizes a perturbation technique to track the operating point that achieves the highest efficiency for the solar panel (Djilali et al., 2024). The system controls the charging current or voltage at specific time intervals and measures the resulting power to determine whether this power has increased or decreased. This approach relies on regulating variables to ensure system stability and achieve maximum efficiency in converting SE into EE. Fig. 4 depicts a flowchart of P&O control.

To highlight the control-related issues with the conventional P&O algorithm, we subject the PV system to rapid load changes. This rapid load variation poses a significant challenge to solar energy systems because it can reduce system effectiveness and efficiency. By understanding how these rapid load changes affect system performance, we can develop better control solutions that improve the system response and enhance its efficiency.

Fig. 5a and b depict the curves of rapid resistive load change, which is fundamental to understanding the impact of dynamic load variations on the performance of the PV system.

Fig. 5a depicts the sudden and rapid increase in the resistive load. At 2 s, the resistive load reached 5 ohms, which then rose rapidly to 7 ohms. Subsequently, at 6 s, the value increased to 10 ohms. This sudden change in resistive load poses a significant challenge to SE systems, leading to fluctuations in power output and impacting system stability, as depicted in Fig. 6a. Figs. 6b and 6c show the power loss due to the sudden increase in the resistive load.

Therefore, understanding the system's response to these rapid resistive load changes is crucial for developing conventional P&O control strategies to overcome these challenges and ensure optimal system performance.

Additionally, Fig. 5b depicts the sudden and rapid decrease in the resistive load. At 3 s, the resistance was 10 ohms, which then decreased rapidly to 5 ohms. Finally, at 7 s, the value decreased to 3 ohms. Similar to rapid resistive load increases, rapid resistive load decreases also pose challenges to SE systems, affecting system efficiency and stability, as shown in Fig. 7. Thus, improving conventional P&O-MPPT control performance is necessary to effectively manage and adapt to these dynamic load variations.

Fig. 8 shows the Duty cycle pattern emerging from conventional P&O control during rapid load changes. Through the analysis of the curves in Figs. 8b, 8c, 8e, and 8f, a clear drift behavior is observed. Instead of swiftly adapting to rapid load changes, P&O control takes a considerable amount of time ($t=0.025$ s, $t=0.049$ s, and $t=0.037$ s) to return to the

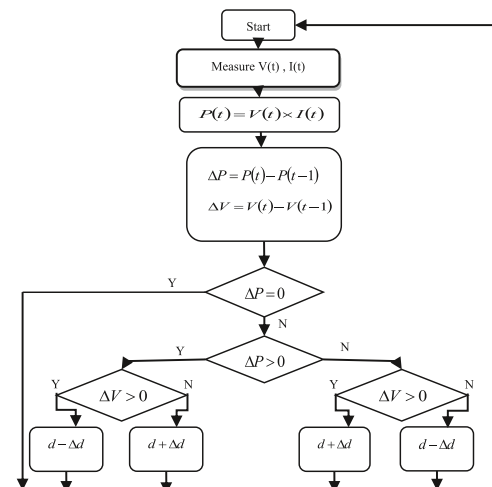


Fig. 4. P&O control.

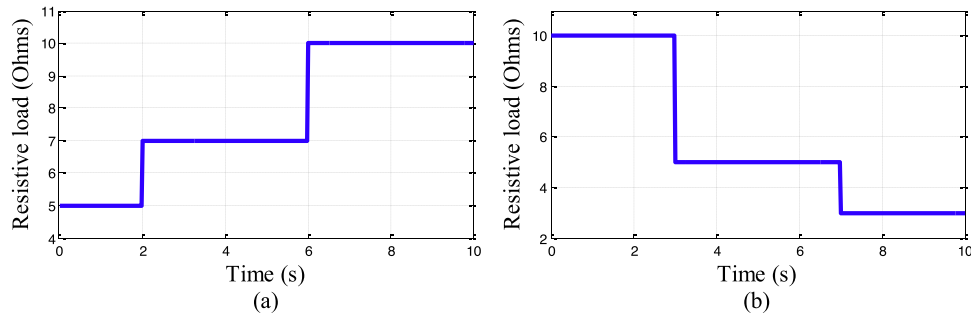


Fig. 5. Curves of rapid resistive load change.

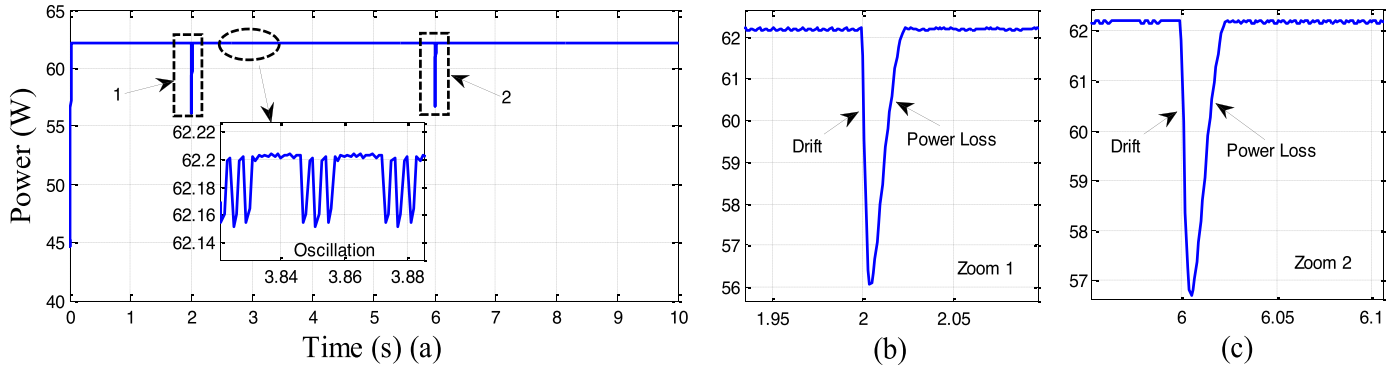


Fig. 6. Sudden resistive load increase.

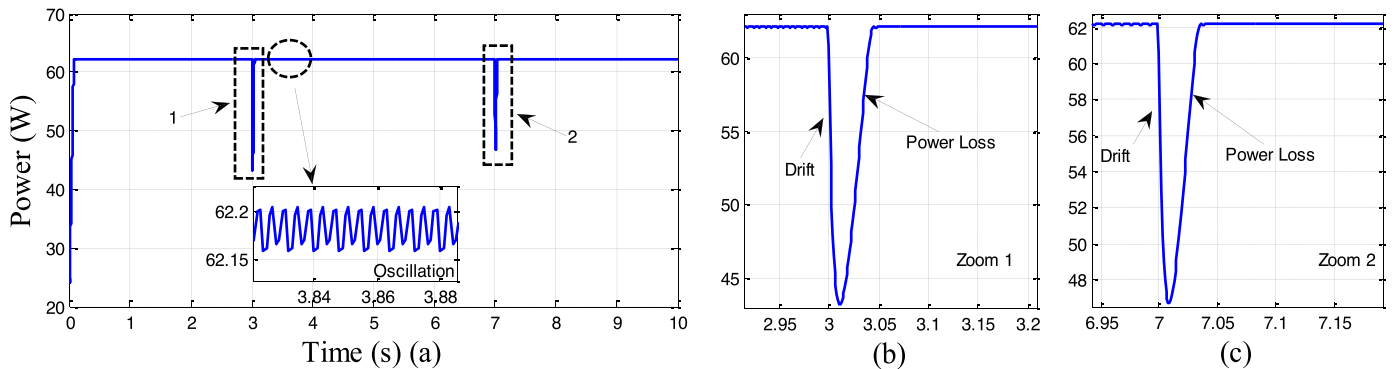


Fig. 7. Sudden resistive load decrease.

MPP, resulting in significant power loss. This drift reflects the inefficiency of the system in handling dynamic load changes. Therefore, this issue presents a technical challenge that requires innovative solutions to enhance conventional P&O-MPPT technique performance to reduce power loss.

Fig. 8a shows that the duty cycle gradually increases from 0.462 to 0.512 (Fig. 8b), forming a sloping staircase curve, as the rapid resistive load increases from 5 ohms to 7 ohms. Similarly, it increases slowly from 0.506 to 0.556 (Fig. 8c) when the resistive load increases rapidly from 5 ohms to 10 ohms. Conversely, during a rapid resistive load decrease, the duty cycle also decreases gradually (Fig. 8d). For instance, when the resistive load quickly decreases from 10 ohms to 5 ohms at 3 s, the duty cycle decreases slowly from 0.556 to 0.458 (Fig. 8e). Similarly, it decreases slowly from 0.47 to 0.396 (Fig. 8f) when the resistive load rapidly decreases from 5 ohms to 3 ohms. These gradual changes in the duty cycle during rapid resistive load variations contribute to the formation of sloping staircase curves. This phenomenon requires further study and understanding to improve system performance and reduce

energy loss.

2.4. Drift and power loss problems

In this section, we analyze how drift and power loss occur due to rapid changes in the resistive load on the power-voltage curve. A profound understanding of this phenomenon is crucial for improving the performance of P&O-MPPT control in facing challenges related to rapid variations in resistive load. When a sudden increase in the resistive load occurs, as shown in Fig. 9, the MPPT technique swiftly moves to the right as the P&O control attempts to adapt to this change by gradually increasing the duty cycle. However, due to the sluggish nature of the duty cycle, returning to the MPPT strategy requires a considerable amount of time. Similarly, when a sudden decrease in the resistive load occurs, the MPPT strategy shifts to the left. This rapid movement results in power loss as the P&O control system attempts to adjust by reducing the duty cycle. A gain, due to the slow response of the duty cycle, the system takes time to return to the MPPT technique and restore stability.

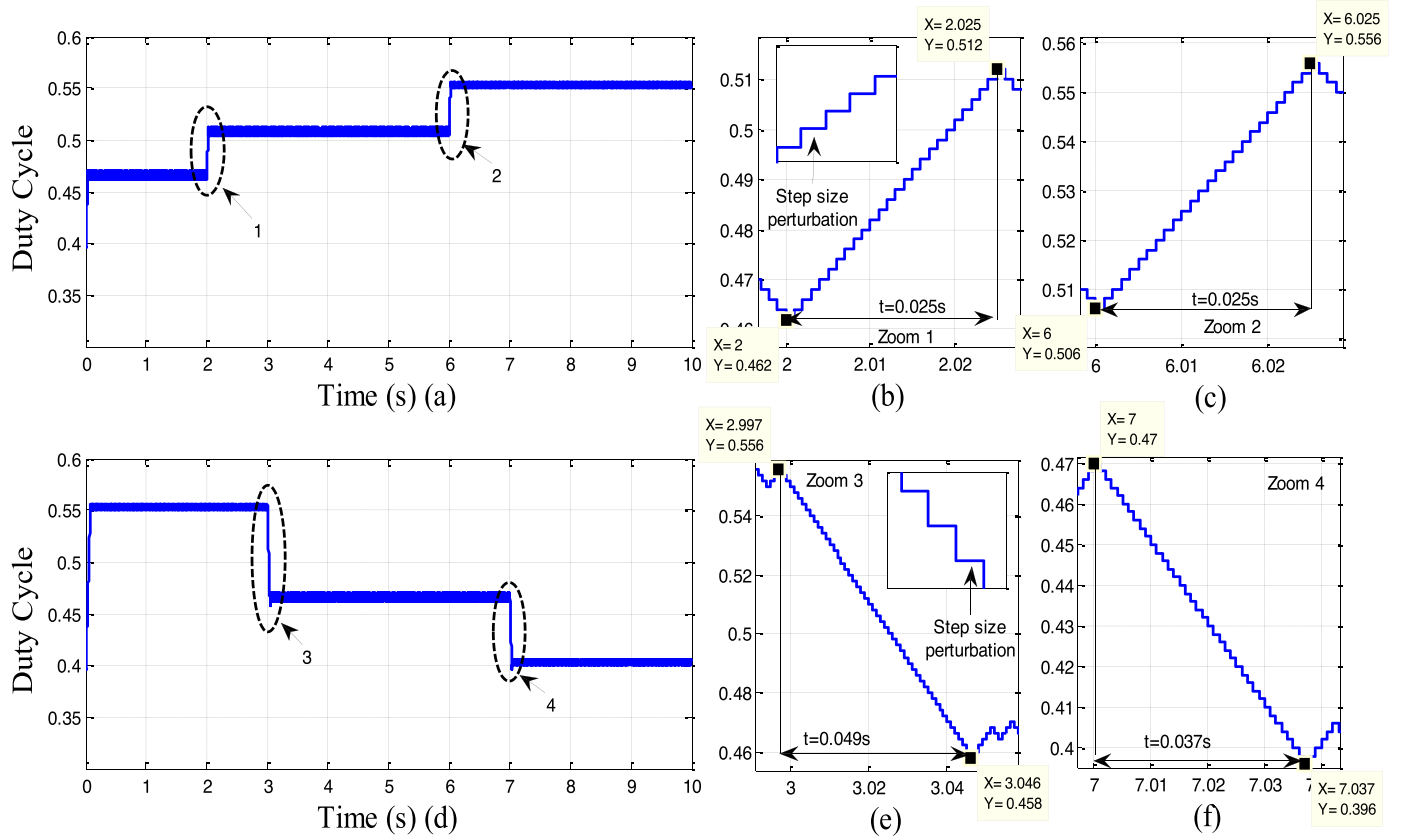


Fig. 8. Duty cycle in sudden resistive load change.

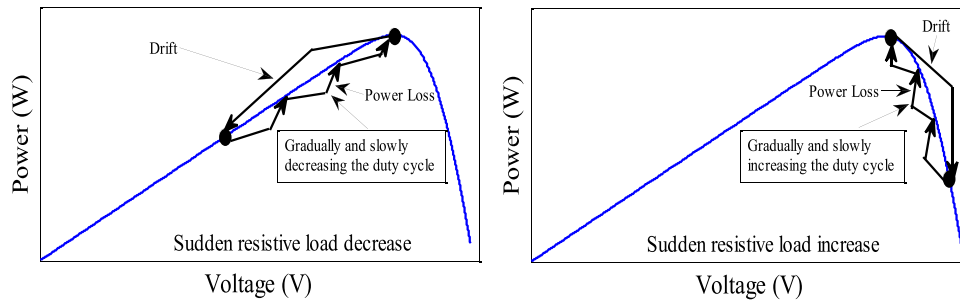


Fig. 9. Drift and power loss problems.

Notably, the primary cause of this drift and power loss is the delayed response of the sluggish duty cycle. Therefore, significant emphasis is placed on improving the performance of the duty cycle to address this delay and enhance the stability of the PV system under such changing conditions.

3. Proposed solution to address the problem of oscillation

To address the issue of oscillation around the MPPT technique, we employed the "step size perturbation" method, which has been utilized in (Djilali et al., 2024) and is characterized by the following equation:

$$D(t) = D(t-1) \pm M \times \left| \frac{\Delta P}{\Delta V} \right| \quad (14)$$

The parameter $D(t)$ denotes the duty cycle of the power converter at time t .

Through Eq. (14), we can obtain the following:

$$D(t) = D(t-1) \pm M \times \left| \frac{\Delta(V \times I)}{\Delta V} \right| \quad (15)$$

Haut du formulaire

Upon calculating the derivative, the resulting equation is as follows:

$$D(t) = D(t-1) \pm M \times \left| \frac{\Delta I}{\Delta V} + \frac{I}{V} \right| \quad (16)$$

The PV output current and the slope of the absolute value of the derivative of the current with respect to the voltage plus current divided by the voltage curves can be obtained accordingly, as depicted in Fig. 10.

The curves representing the (Current-Voltage) and $[Abs(\Delta I/\Delta V + I/V)]-V$ relationships are presented in Fig. 10. Fig. 10 reveals that the duty cycle D employed in the VSS perturbation P&O-MPPT method exhibits dynamic behavior. Specifically, when the power point is in proximity to the MPPT strategy, the MPPT control applies a small step size perturbation. Conversely, when the power point coincides with the MPP, the MPPT control sets the step size to zero. The operational principle of the

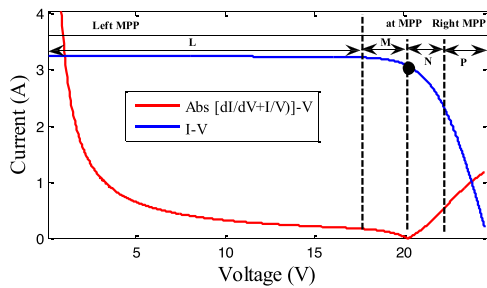


Fig. 10. Current-voltage and $[Abs(dI/dV+I/V)]$ -V curves.

VSS perturbation is delineated in Table 1. This depiction in Fig. 10 and the elaboration in Table 1 offer a comprehensive understanding of the dynamic nature of the duty cycle in the VSS perturbation P&O-MPPT method. This finding underscores the adaptability of the MPPT control strategy in response to varying operating conditions, ultimately enhancing the efficiency and effectiveness of PV systems.

4. Proposed solution to address the problem of drift

To overcome the problem of drift during rapid resistive load changes and therefore to avoid power loss, an effective and impactful method consisting of two steps is proposed:

First step: When a rapid change in the resistive load occurs, a command is issued to prevent the algorithm from functioning (P&O in the off state). This approach aims to prevent drift resulting from the emergence of a large step size perturbation due to incorrect information reading.

Second step: This step involves taking immediate and effective actions when sudden and rapid changes in the resistive load occur. This step is executed by applying the duty cycle outlined in the following equation:

$$D(t) = D(t-1) \pm K \times D_F \quad (15)$$

where D_F represents a fixed step size and K is greater than M (K and M are unchanging constant parameters).

Through this method, we achieve maximum efficiency and sustainability in energy use. This approach helps to avoid drift and energy loss during rapid changes in the load, thereby enhancing system efficiency and reducing the resulting loss.

To capture rapid changes in the resistive load, we consider $|dR_L/dt|$ and ϕ , where:

$$\phi = \frac{1}{dt} \quad (16)$$

dt : Simulation step

When $|dR_L/dt| \geq \phi$, this indicates the presence of a rapid change in the resistive load, and thus:

1. If $dR_L > 0$ increases the duty cycle $[D_F(t)]$ by $+K \times D_F$
2. If $dR_L < 0$ decreases the duty cycle $[D_F(t)]$ by $-K \times D_F$

When $|dR_L/dt| < \phi$ and $dR_L = 0$ in this case, the modified P&O control operates with the proposed duty cycle outlined in Eq. (16).

Fig. 11 shows the benefit of function dR_L/dt in capturing sudden and rapid changes in the resistive load.

Table 1
Variable step size P&O principle.

Location	Left of MPP		At MPP	Right of MPP	
Interval	L	M	0	N	P
$Abs(dI/dV+I/V)$	Large	Small	0	Small	Large

Fig. 11b clearly shows that dR_L/dt quickly and effectively captures any sudden or rapid changes in the resistive load. Comparing Fig. 11a and 11b reveals that at 2 s, 3 s, 6 s, and 7 s, when rapid changes occurred, the function of dR_L/dt successfully swiftly captured these changes. This capability allows for the rapid mitigation of drift and power loss via the proposed modified P&O-MPPT control. The proposed modified P&O control scheme was implemented in MATLAB/Simulink, as depicted in Fig. 12.

5. Results

In this section, modified P&O control is implemented using the MATLAB environment, where several different tests are proposed to study the performance and effectiveness of the proposed strategy in improving the characteristics of the studied energy system compared to the traditional strategy. The system parameters used in this work are shown in Table A1 (see Appendix).

5.1. Effectiveness of proposed modified P&O control in rapid changes in resistive load

Fig. 13 shows that the modified P&O control exhibits a significantly improved performance compared to the traditional P&O control in dealing with drift and oscillation issues. This improvement is attributed to modifications in the duty cycle, making it more rapid and effective in responding to rapid changes in resistive load. Additionally, Fig. 13b, c, e, and f show a clear change in the behavior of the duty cycle from a slow inclined shape to a fast vertical shape. This change facilitates the avoidance of drift and reduces power loss, thereby enhancing system efficiency.

Furthermore, Fig. 13 demonstrates the ability to overcome oscillation by adjusting the perturbation step size based on the MPPT technique. This adjustment improves the system's stability and reduces oscillations, thus enhancing the overall system performance.

Based on these data, the modified control scheme exhibits greater effectiveness than does the traditional approach in addressing drift and oscillation issues because of the new modifications introduced to improve its responsiveness and overall system performance.

Fig. 14 clearly shows that the ability of the modified P&O control to eliminate drift and oscillation, which results in power loss, is in stark contrast to traditional P&O control. Fig. 14 clearly shows that the modified duty cycle effectively contributes to avoiding drift during rapid changes in the resistive load, both during increases and decreases. Fig. 14 shows that the modified P&O control can quickly respond to changes in the resistive load, thus enhancing the system efficiency and reducing the power loss. In contrast, traditional P&O control struggles with rapid changes, resulting in undesirable oscillations and significant power loss. Furthermore, Fig. 14 demonstrates that the modified P&O control can maintain system stability without causing oscillations, thus improving system efficiency and reducing energy costs. Fig. 14 highlights significant performance differences between the modified P&O control and traditional control in avoiding drift and oscillation, making it a preferred option for applications requiring stability and high energy efficiency.

5.2. Effectiveness of proposed modified P&O control under rapid changes in solar irradiation

The results of Fig. 15 demonstrate the superior capability of the modified P&O technique to accurately track the MPP with zero oscillations around P_{MAX} and with minimal power loss during rapid changes in solar irradiance. Fig. 15a and d illustrates a sudden increase in solar irradiance from 200 W/m^2 to 1000 W/m^2 , followed by a sudden decrease to 200 W/m^2 , reflecting the system's exceptional ability to quickly adapt to changes in solar irradiation.

These results bolster confidence in the application of the modified

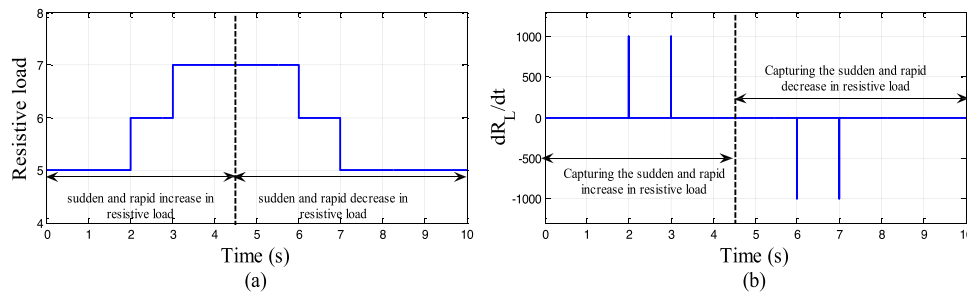
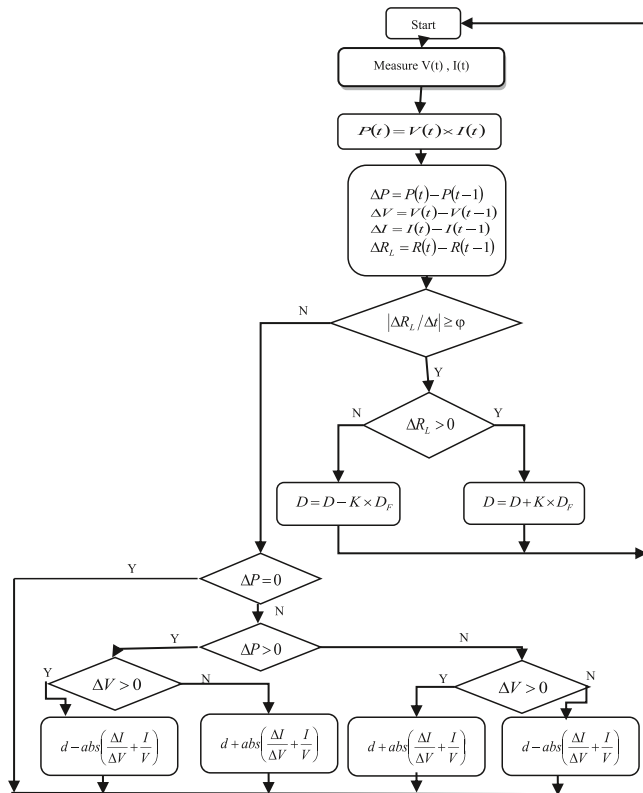
Fig. 11. dR_L/dt curve.

Fig. 12. Proposed modified P&O control.

P&O technique in SE generation systems, paving the way for its widespread use in the future to enhance the performance and efficiency of these systems. It is noteworthy that this modified MPPT method can contribute to enhancing energy sustainability and environmental protection by increasing the efficiency of SE generation and reducing reliance on other energy sources that cause pollution.

These findings also underscore the importance of research and development in MPPT techniques in SE systems, as these techniques can enhance the efficiency and sustainability of SE as a clean and efficient alternative to traditional energy sources.

5.3. Effectiveness of proposed modified P&O control under rapid changes in temperature

A study tracking the MPP of the PV system under temperature variations is an important topic in the field of SE generation. Fig. 16 illustrates the rapid change of temperature in the PV system, increasing from 25 degrees Celsius to 29 degrees Celsius, and then decreasing back to 25 degrees Celsius. Fig. 16a provides a detailed insight into the temperature changes.

In Fig. 16b, the direct correlation between temperature and generated power from the photovoltaic system is demonstrated, where an increase in temperature leads to a decrease in generated power, and vice versa. This effect, demonstrated in Fig. 16b, highlights the role of temperature as a significant factor to be considered in the design and operation of SE generation systems.

Furthermore, Fig. 16b and the zoom (Fig. 16c) illustrate how the modified P&O method can be effective in tracking the MPP of the system under temperature variations without oscillation around the MPP.

One of the key differences between the modified P&O method and the traditional one is the method used to determine the MPP. While the modified P&O relies on modifications and enhancements to increase its accuracy and efficiency, the traditional P&O relies on conventional methods without applying any additional modifications.

The results presented in the simulation indicate that the modified P&O method outperforms the traditional method in tracking the MPP of the PV system, enhancing its use in applications requiring rapid response and high accuracy amidst various environmental changes.

In Table 2, a comparison is made between the traditional P&O-MPPT technique and the proposed modified P&O-MPPT strategy in terms of response time in the case of rapid change in solar irradiation, rapid change in temperature, and rapid change in resistive load. The proposed modified P&O-MPPT strategy provided better response time in the case of increase and decrease than the traditional P&O-MPPT strategy, which is a positive thing that indicates the extent of the high performance that characterizes this strategy. In the case of rapid change in solar irradiation, the proposed modified P&O-MPPT strategy provided a better time than the traditional P&O-MPPT strategy, as this reduction was estimated at percentages of 36.36 % and 70.58 % for increase and decrease, respectively. Also, the proposed modified P&O-MPPT strategy reduced the response time in the case of rapid change in temperature by percentages estimated at 72.72 % and 50 % for both increase and decrease, respectively, compared to the traditional P&O-MPPT strategy. In the case of rapid change in resistive load, the proposed modified P&O-MPPT strategy provided an unsatisfactory response time compared to the traditional P&O-MPPT strategy when it increased, which is an undesirable negative. Therefore, the traditional strategy reduced this time in this case by an estimated percentage of 72.50 % compared to the proposed modified P&O-MPPT strategy. However, the proposed modified P&O-MPPT strategy gave a better time than the traditional P&O-MPPT strategy in the case of a decrease, as the response time was reduced by an estimated percentage of 66.66 % compared to the traditional P&O-MPPT strategy. This presented ratio indicates the extent of the efficiency and effectiveness of the proposed modified P&O-MPPT strategy in improving the dynamic characteristics of the studied energy system compared to the traditional P&O-MPPT strategy.

In Table 3, a comparison is made between different papers and the proposed modified P&O-MPPT method in terms of Steady-state oscillation, Efficiency, and Sensors. Also, the MPPT strategy adopted to control the PV system is mentioned. This looks at the efficiency and ability of the proposed modified P&O-MPPT strategy in improving the characteristics of the PV system compared to several other strategies that have been

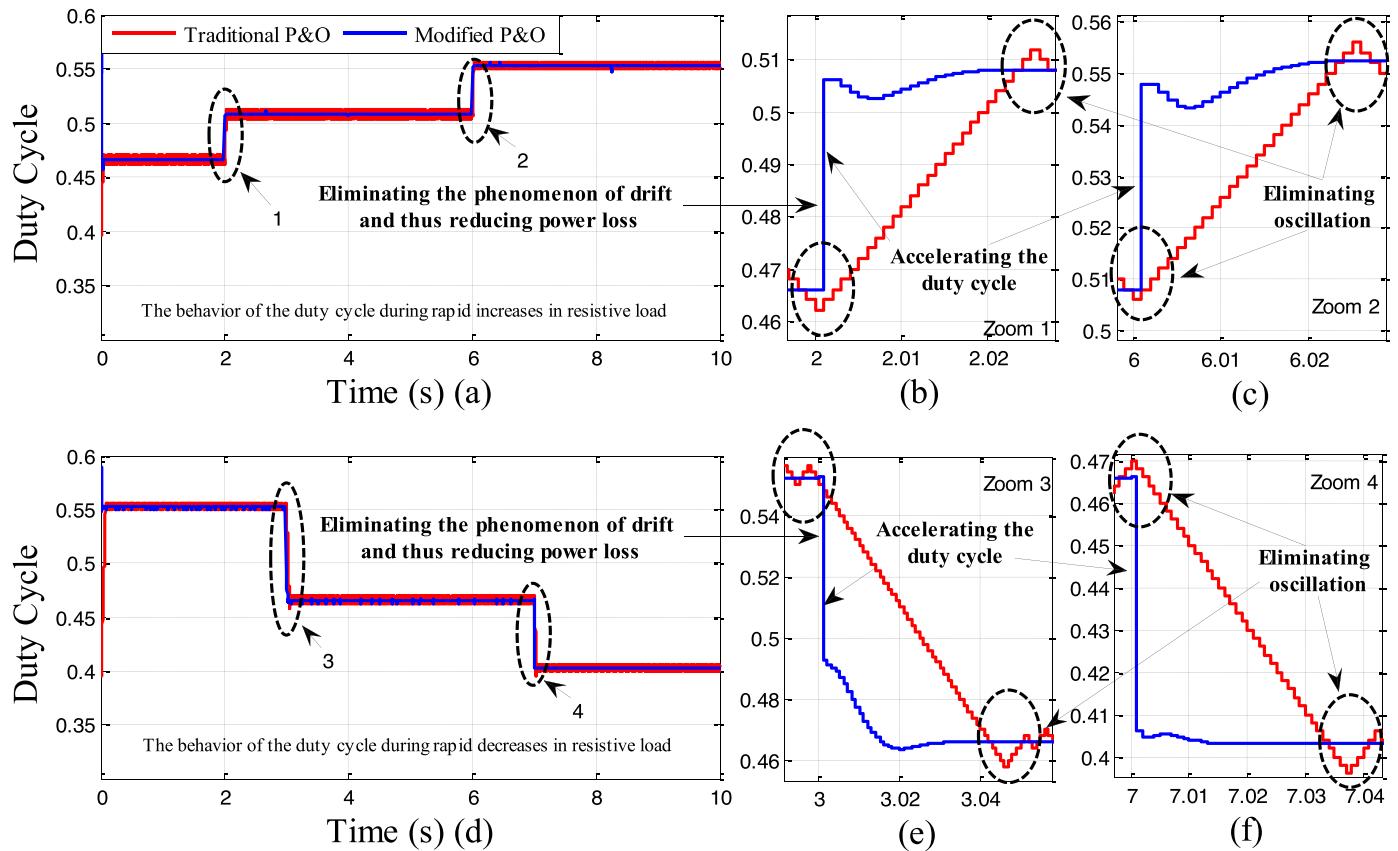


Fig. 13. Comparing the behavior of the work cycle for traditional P&O and modified P&O during rapid changes in resistive load.

proposed in several scientific papers. Therefore, the proposed modified P&O-MPPT strategy has high efficiency compared to several controls, which is a positive thing that makes the proposed modified P&O-MPPT strategy an appropriate solution in the future. The proposed modified P&O-MPPT strategy in terms of steady-state oscillation is better than several other controls, which is a good thing that shows the effectiveness and performance of the proposed modified P&O-MPPT strategy. In the proposed modified P&O-MPPT strategy, voltage, current, power, and irradiation sensors were used, which makes it different from other works. The latter uses mostly current and voltage sensors. So the proposed modified P&O-MPPT strategy is different from several scientific works, as its reliance on several types of sensors makes it give good results. However, its use of several sensors makes it expensive, which is an undesirable negative.

6. Conclusions

A modified P&O control method has been developed to significantly enhance the effectiveness in addressing drift (power loss) and oscillation issues over traditional P&O control methods. This improvement is attributed to the faster duty cycle, which enables the proposed method to react promptly and efficiently to rapid changes in the resistive load. To mitigate drift and oscillation, the behavior of the duty cycle in the modified P&O control was altered from a slow inclined shape to a fast vertical shape. This change eliminated drift and reduced energy loss, thereby enhancing the overall efficiency of the PV system. The modified control is capable of improving the performance of the PV system by increasing the response speed and reducing drift and oscillation. This makes the proposed method a more effective option for enhancing the technical performance of the PV system and providing cost savings by preventing power losses.

In the future, we will try to carry out this work experimentally and

verify the results obtained by comparing them with other experimental works. Also, it is proposed to use new strategies to increase the durability and performance of the MPPT strategy of PV system, where it will be proposed to combine fractional calculus, grey wolf optimization, and synergetic control to obtain a new control characterized by high performance.

7. Institutional review board statement

Not applicable.

8. Informed consent statement

Not applicable.

9. Funding

Not applicable.

10. CRediT authorship contribution statement

Abdelkadir Belhadj Djilali: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dalal Zellouma:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Formal analysis. **Ayman Alhejji:** Writing – original draft, Visualization, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis. **Habib Benbouhenni:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal

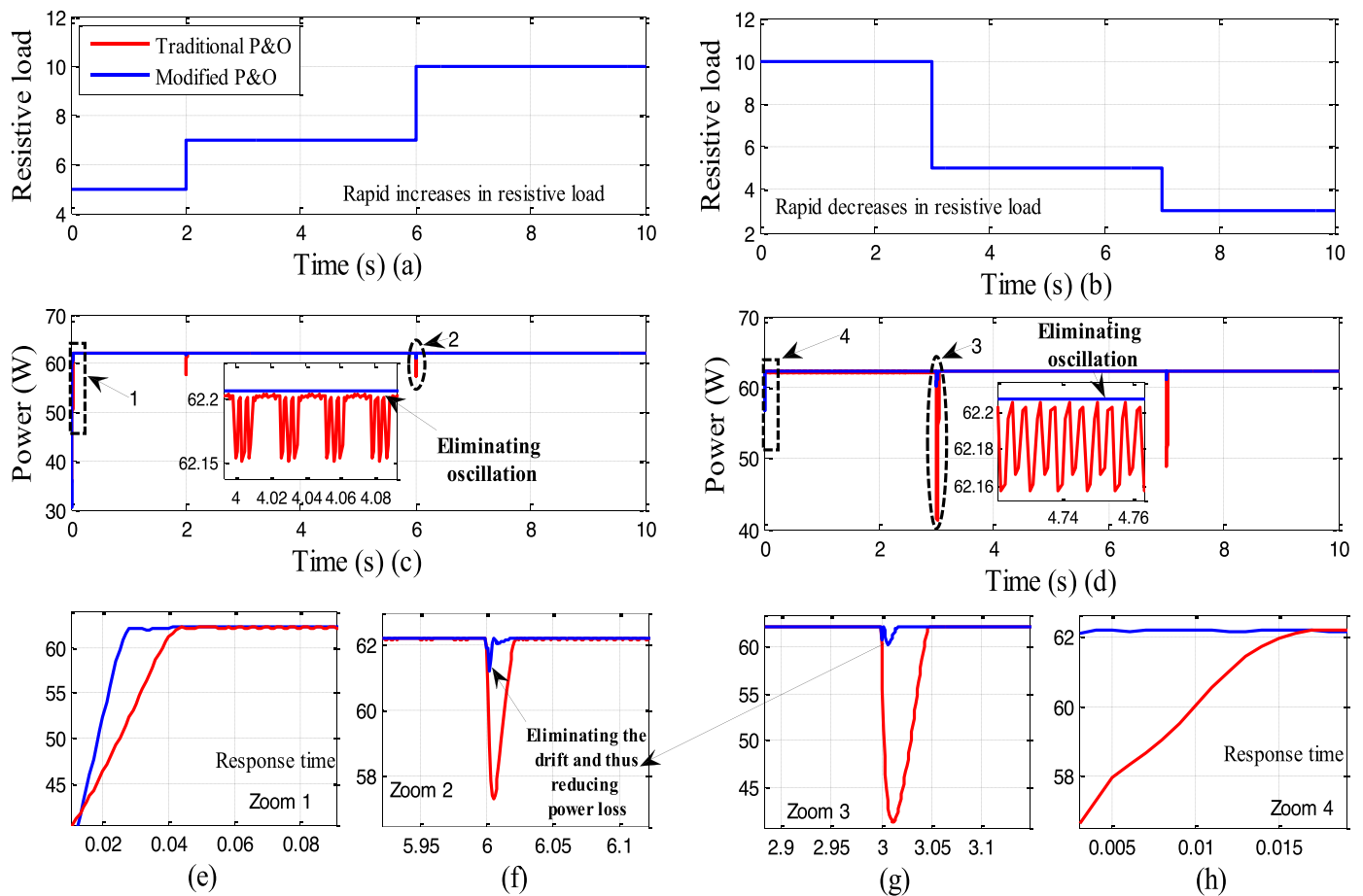


Fig. 14. Comparing the power output of traditional P&O and modified P&O and their ability to eliminate oscillation, drift, and power loss.

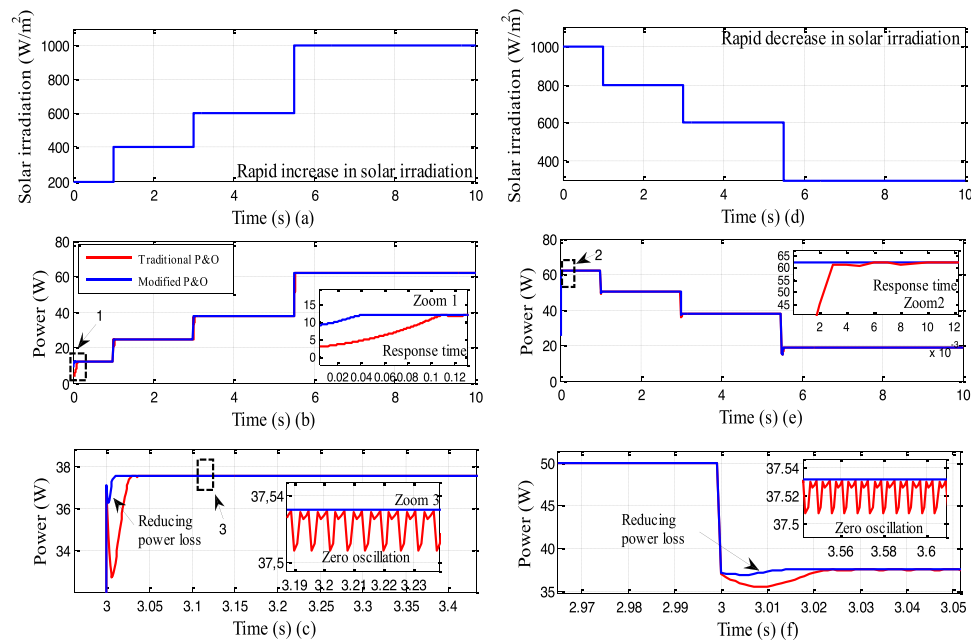


Fig. 15. Comparing the power output of traditional and modified P&O under rapid change in solar irradiation.

analysis, Data curation, Conceptualization. **Adil Yahdou:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis. **Elhadj Bounadja:** Writing – review & editing, Visualization, Supervision, Resources, Methodology,

Investigation, Formal analysis.

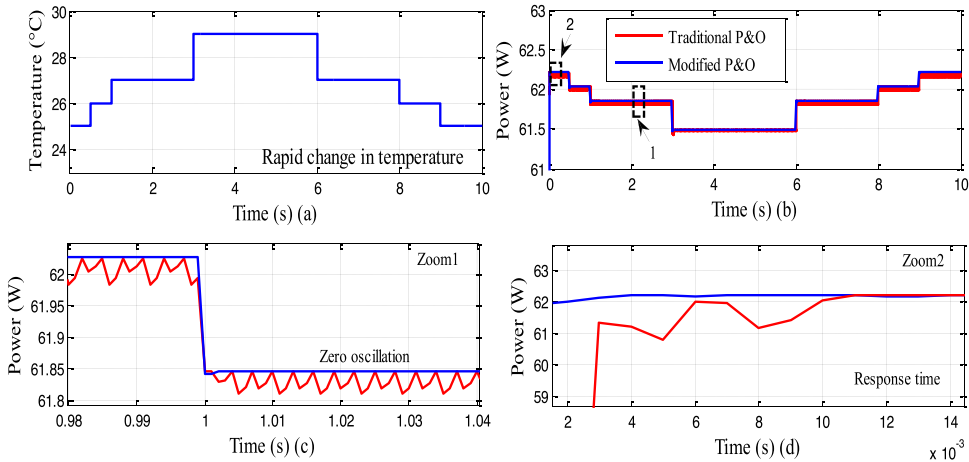


Fig. 16. Comparing the power output of traditional and modified P&O under rapid change in temperature.

Table 2
Response time comparison between traditional and modified P&O-MPPT technique.

MPPT methods	Rapid change in solar irradiation		Rapid change in temperature		Rapid change in resistive load	
	Increase	Decrease	Increase	Decrease	Increase	Decrease
Traditional P&O-MPPT technique	0.044 s	0.017 s	0.011 s	0.018 s	0.011 s	0.003 s
Modified P&O-MPPT technique	0.028 s	0.005 s	0.003 s	0.009 s	0.04 s	0.001 s
Ratios	36.36 %	70.58 %	72.72 %	50 %	72.50 %	66.66 %

Table 3
Comparison between different papers and proposed modified P&O-MPPT method.

Ref	MPPT technique	Efficiency	Sensors	Steady-state oscillation
(Guerra et al., 2021)	Incremental conductance	High	Voltage and Current	Yes
(Li et al., 2016)	Short Circuit current	Low	Current	
	Neural Network	High	User knowledge	
	Open circuit voltage	Low	Voltage	
	Short Circuit current		Current	
	P&O (variable perturbation size)	High	Current and Voltage	No
	P&O (fixed perturbation size)	Low		Yes
(Serrano-Guerrero et al., 2016)	PSO	High		No
(Tang et al., 2017)	Combining fractional-order and FLC			Yes
(Soon and Mekhilef, 2015)	Fast converging technique			
(Tey et al., 2018)	Differential Evolution-Based MPPT			
(Ahmad et al., 2015)	Modified incremental conductance			
Proposed MPPT control			Voltage, current, power	No

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

Appendix

System parameters

In Table A1 the parameters of the system used in this work are given.

Table A1
System parameters

Parameter	Value
I_{SC} (Short circuit current at standard conditions)	3.25 A
R_s (Series-Resistance)	0,015 Ohm
A (Diode ideality factor)	1,2
I_0 (saturation current)	8225.10^{-12} A
q (Electron charge)	$1,6.10^{-19}$ C
K (Boltzmann constant)	$1,38.10^{-23}$ J/K
T_{SC} (Temperature at standard conditions)	25 C°
V_{OC} (Open-Circuit Voltage)	24.8 V
R_{sh} (Parallel-Resistance)	30 Ohm
G (Solar irradiation at standard Conditions)	1000 W/m ²
N_s (Number of solar cells in series)	36
N_p (Number of solar cells in parallel)	1

Proposed modified P&O technique in MATLAB software

Fig. 17 shows the shape taken from the MATLAB environment and used to complete the proposed work. This figure give a clear picture of the proposed MPPT control and the extent to which it differs from several existing works in the scientific field.

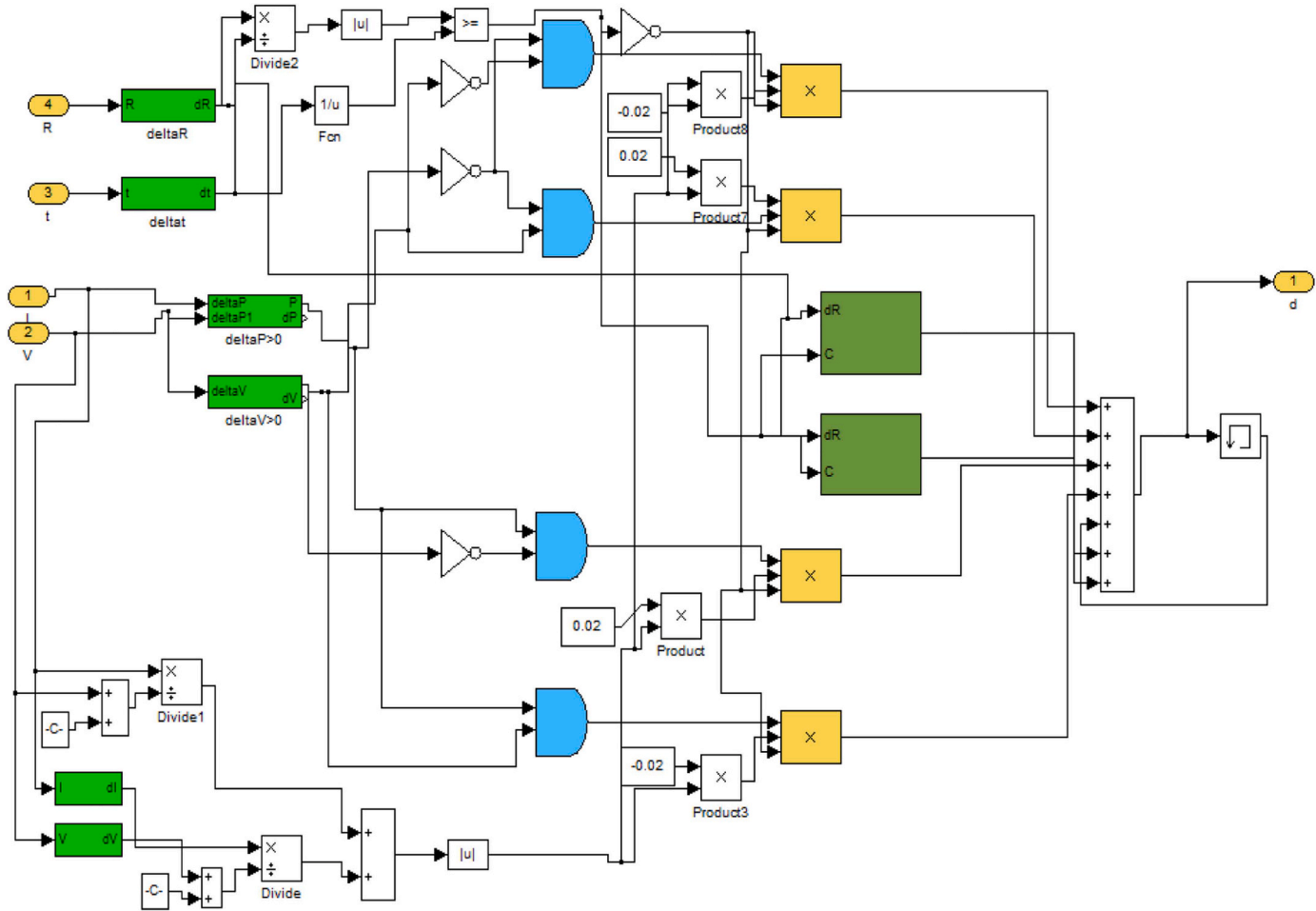


Fig. 17. Proposed modified P&O technique in MATLAB software.

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